D12. Final Technical Report and Demonstration

Consolidated conclusion, deliverable map and 5-10 minute technical demo script.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

What a reviewer can verify

Review question	Where the submission answers it
Why this architecture?	D1 and D4: technology rationale, provenance and hybrid retrieval
Which model/configuration was tested?	D2, D3 and D6: controlled model, ICL and prompt outcomes
Does retrieval actually work?	D5: metrics and evidence-case analysis
Can it be reproduced?	D9 and final_validation ledger: configs, logs and outputs
What does the POC prove and not prove?	D10 and D11: results, errors, gate and limitations

Final submission position

This is a strong technical POC submission because it combines a modern local RAG architecture with a traceable evaluation harness and actual controlled experiments. It tells a coherent story from research rationale to implementation to comparative evidence, instead of treating the model as a black box. [E8][E10]

The final spoken demo should lead with the auditable chain: report -> provenance-preserving evidence -> hybrid retrieval -> constrained answer -> question-level record -> audit score -> release gate. Finish by stating the precise next requirement: independent expert labels across more reports. [E4][E5][E10]

Final conclusion

The submitted system is a local, evidence-traceable proof of concept for benchmarking individual cybersecurity audit reports. It can parse supplied JSON reports, retrieve source evidence, produce structured Yes/No/Insufficient Evidence decisions, retain question-level outputs and compute an audit-level score. [E4][E5][E8]

The strongest measured real-report result is 81.25% accuracy across 16 source-derived question/report pairs, with Recall@8 of 87.50%, Precision@8 of 14.43%, MRR of 0.2781, nDCG@8 of 0.4212, 100% implemented grounding rate and 13.66-second mean latency. [E5]

These results are implementation/integration evidence only. The dataset has two reports, 12 Yes labels, four No labels and zero Insufficient Evidence labels; therefore it cannot support a production or generalisation claim. [E5]

Deliverable status

Work package	Delivered evidence	Status
WP1: LLM and ICL	D1-D3: candidate review, criteria, ICL implementations and controlled comparison plan	Implemented; comparative ICL/model evidence incomplete [E2][E3]
WP2: Retrieval and chatbot	D4-D7: parser, chunker, hybrid index, prompt, pipeline and offline check	Functional proof of concept [E7][E8]
WP3: Evaluation	D8-D11: labelled questions, harness, artifacts, metrics, error analysis and two-report POC	Measured integration evidence; release gate blocked [E4][E5]

Recommended 5-10 minute demo

Time	Demonstration step	Evidence to show
0:00-1:00	State objective, benchmark scoring flow and air-gapped constraint	This report and D1
1:00-2:30	Show parser, structured chunks and hybrid retrieval design	D4; src module map [E8]
2:30-4:00	Show v2 evidence-first prompt and one question-level record	D6; question_level_results.json [E5]
4:00-5:30	Show two-report metrics and audit scores	D10; metric and score JSON [E5]
5:30-7:00	Run or explain offline mock verification	D7/D9; 28 Aug test record [E7]
7:00-9:00	Explain limitations, release gate and next experiment	D8-D10; blocked evidence boundary [E5]

Next work justified by the results

First, independently label a report-level held-out set with enough No and Insufficient Evidence cases. Second, run controlled ablations of ICL, chunking, hybrid retrieval, reranking and CRAG. Third, evaluate claim entailment, contradictions, OCR noise and visual evidence if that input scope is accepted. [E5][E8]

Do not add late interaction, graph or visual complexity by default. Each should be selected only if it improves the frozen evaluation under its intended question type. [S2][S3][S6]

Evidence and external sources

The following repository artifacts and literature sources substantiate this report. No external source is used to claim an unmeasured project result.

Final review claims and support

Final claim	Supporting evidence	Boundary
A local RAG pipeline was implemented	Parser, hybrid index, prompt, harness and 16-pair run. [E4][E8]	Text/JSON scope; visual understanding unvalidated.
Architecture choices were tested	LLM, ICL, prompt, retrieval, CRAG and chunking comparisons. [E10]	Small source-derived dataset.
Scores are traceable	Question records, metrics, audit scores, logs and config. [E4][E5]	Gold labels are not independent expert labels.
The POC is technically credible	Two reports completed with zero execution errors. [E4]	Not a production readiness claim.

Submission conclusion

This final package demonstrates the complete requested workflow: audit report input, benchmark question, evidence retrieval, constrained local LLM response, positive/negative assessment, audit score and benchmark decision. Its central contribution is traceability: every claimed score has a route back to source report, retrieval evidence, configuration and answer record. [E4][E5]

The final recommendation is to submit this as an evaluated, auditable proof of concept. State plainly in the review that broader deployment requires independently labelled multi-report validation, balanced answer classes and visual/OCR coverage where the input data demands it. [E10]

Repository evidence cited

- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.
- [E5] results/production_real_reports/1787846335_de0ded86a7db/{retrieval_metrics,question_level_llm_metrics,sota_faithfulness_metrics,audit_level_scores}.json: scored metrics and audit decisions.

- [E8] src/{docling_parser.py,chunking.py,embed_index.py,pipeline.py,evaluation_harness.py,adaptive_rag.py,prompts.py,icl.py}: implementation inspection.
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.

External technology sources

- [S1] Chen et al. (2024), BGE-M3-Embedding, arXiv:2402.03216, https://arxiv.org/abs/2402.03216.
- [S2] Santhanam et al. (2021), ColBERTv2, arXiv:2112.01488, https://arxiv.org/abs/2112.01488.
- [S3] Faysse et al. (2025), ColPali, arXiv:2407.01449, https://arxiv.org/abs/2407.01449.
- [S4] Yang et al. (2025), Qwen3 Technical Report, arXiv:2505.09388, https://arxiv.org/abs/2505.09388.
- [S5] Qwen (2025), Qwen3-Reranker-4B model card, https://huggingface.co/Qwen/Qwen3-Reranker-4B.
- [S6] Microsoft GraphRAG query documentation, https://microsoft.github.io/graphrag/query/overview/.